

Mikael Ögren

The Design of an Array Microphone

Abstract

The theory of array microphones has been studied and summed up in a compact way. The concepts of weighting, spacing and constant beam width have been explained, and a method to focus arrays through time delays has been proposed.

A cheap PC based array microphone system for frequencies below 4 kHz, from preamplifiers to analysis programs, has been developed and tested. Measurements indicate that the theoretical predictions of beam width and signal to noise ratio can be achieved with this system. The performance in terms of beam width, has been compared with that of a parabolic reflector with diameter 2 m, and the results show that the array system has a better performance at low frequencies than the parabola.

Key words: microphone, array, acoustics, acoustic measurements

SP

SP Report 1996:38
ISBN 91-7848-643-2
ISSN 0284-5172
Borås 1996

Swedish National Testing and Research Institute

SP Report 1996:38

Postal address:

Box 857, S-501 15 BORÅS,
Sweden
Telephone +46 33 16 50 00
Telex 36252 Testing S
Telefax +46 33 13 55 02

Contents

Abstract	2
Contents	3
Preface	4
Conclusions	5
1 Theoretical analysis of array properties	7
1.1 Basic array properties	7
1.1.1 Linear array with uniform amplitude and spacing	7
1.1.2 Linear array with progressive phase	9
1.2 Weighting	10
1.2.1 The Binomial array	11
1.2.2 Dolph array microphone	11
1.3 Microphone spacing	12
1.3.1 A simple non equidistant array microphone	13
1.4 Constant beam width	14
1.4.1 van der Waal constant beam width array microphone	14
1.4.1.1 van der Waal spacing	15
1.4.1.2 van der Waal linear phase FIR filters	15
1.5 Focusing the array microphone	16
1.6 Effects of phase and amplitude distortion	18
1.7 Two-dimensional array microphones	19
2 Implementation and evaluation of a few simple array microphones	22
2.1 Initial measurements	22
2.1.1 The two microphone array	22
2.2 An eight element linear array with uniform amplitude and spacing	23
2.2.1 Analog summing amplifiers	23
2.2.2 Measurement in hemi-anechoic chamber	24
2.2.3 Measurement of passing X2 train at Torp	25
3 Design and evaluation of a complete array microphone system	26
3.1 The measurement system	26
3.1.1 Microphones, anti-alias filters and pre-amplifiers	26
3.1.2 Analysis programs in MATLAB code	27
Correction for non simultaneous sampling	28
Bandpass filtering	29
Constant beam width	30
Focusing	30
3.2 Initial measurements with simplified setup	30
3.3 Comparison between the array microphone and a parabola	32
References	35
Annex A, Measurement in hemi-anechoic chamber	36
Annex B, Measurement in hemi-anechoic chamber	39
Annex C, Calculation of the array function	42

Preface

This project started as a diploma work for the acoustics laboratory of SP, under the guidance of Mendel Kleiner at the Department of Applied Acoustics, Chalmers University of Technology and Hans Jonasson, SP. The initial idea was to construct and use an array microphone to characterize the noise sources of different Swedish train types. During the work, the project shifted more to the construction and design of the array system.

I would like to express my gratitude to Tomas Ström, SP, for his skillful help during field measurements and good ideas and help with the hardware parts of the array. Per Bankvall has also contributed, as a part of his summer job, by testing microphones for the array.

Mikael Ögren

Conclusions

- Beam width can be exchanged for signal to noise ratio by individual microphone channel weighting.
- Signal to noise ratio can be exchanged for bandwidth by non-uniform microphone spacing.
- Constant beam width over a frequency band can be achieved at the cost of lobe width.
- The microphones of an array do not have to be very closely matched to keep the basic array properties intact.
- Two-dimensional array microphones can be constructed with the simple linear array as a basic element.
- It is possible to correct for a non simultaneous sampling of the array microphone channels by the use of signal processing.
- The array system presented in chapter 3 can easily beat the performance of a 2 m parabola at low frequencies.

1 Theoretical analysis of array properties

The array theory is the theory of how to use the signals of multiple transducers to produce desired effects. Note that this chapter is very much inspired from antenna theory, and the results are often taken from the theory of electromagnetic fields. This text assumes that the elements of the array are microphones, but they could also be sources. There is strong reciprocity between source and receiver, but with a speaker as the source you often have a strong directivity as further complication. Microphones, on the contrary, can be constructed with good omni-directional characteristics within a reasonable frequency band. The most common two applications are to find a source in space and to aim the measurement to suppress sources other than the primary one. The latter is the application to which this text is concerned.

1.1 Basic array properties

This chapter deals with the basic array properties, such as directivity, upper and lower frequency limit and main lobe aiming.

1.1.1 Linear array with uniform amplitude and spacing

The simplest possible array microphone is the linear array with uniform amplitude and spacing. Linear means that all microphones are placed on a line in space, uniform spacing that the distance d is constant and uniform amplitude that no weighting is applied to the microphone signals.

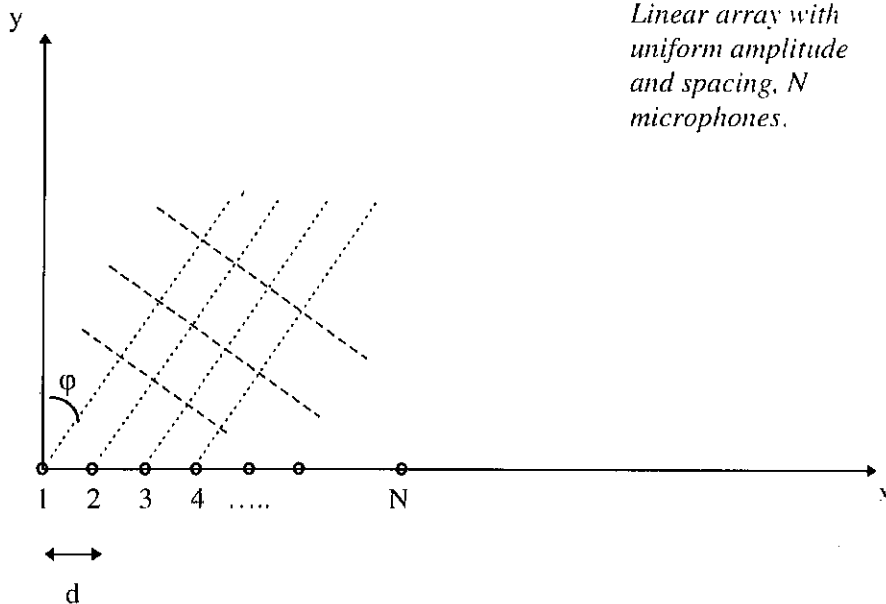


Figure 1.1
Linear array with
uniform amplitude
and spacing, N
microphones.

Assume that we have a plane wave with the angle of incidence φ , and the wavenumber $k=\omega/c$. We analyze by summing the signals from the N microphones. The uniform spacing is d . If the propagating wave has the amplitude 1, our summing amplifier will yield, omitting the factor $e^{-j\omega t}$,

$$F = 1 + e^{jkd \sin(\varphi)} + e^{j2kd \sin(\varphi)} + \dots + e^{j(N-1)kd \sin(\varphi)} = \sum_{n=1}^N e^{j(n-1)kd \sin(\varphi)} \quad (1.1)$$

F is a sum of complex numbers, $|F(\varphi)|$ can assume all values between 0 and N . F is usually called the Array Function, and normalized with the factor $1/N$.

As an example I choose $N=8$ and $d=714$ mm, which gives a total length of 5m. If we calculate $|F(\varphi)|$ for the frequencies 300 and 600 Hz, we get the result displayed in Figure 1.2. $|F(\varphi)|$ is normalized to 60 dB.

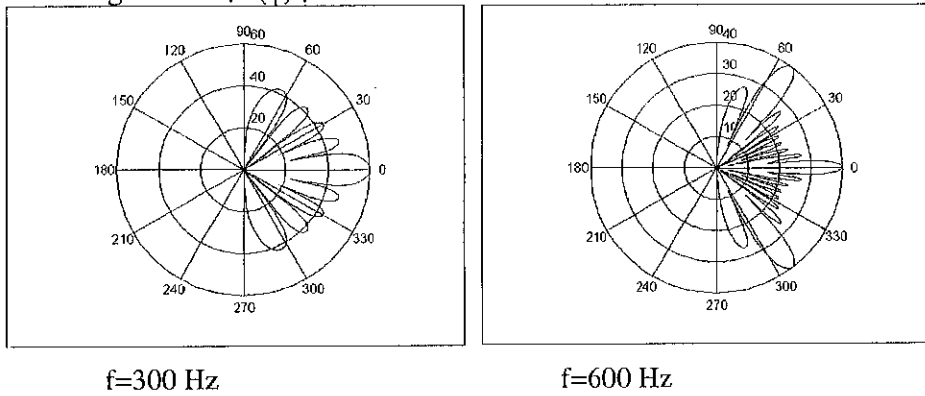


Figure 1.2
 $20\log |F|$ for
 $N=8$ and
 $d=714$ mm
 Normalized to
 60dB.

First we consider $f=300$ Hz. If the angle of incidence is within $\varphi=\pm 10^\circ$, we can clearly see that the signal $|F|$ is amplified. If it falls outside this angle, $|F|$ is attenuated with about 15 dB. For $f=600$ Hz, we see that not only the main lobe, but also the so called grating lobes, are amplified. This makes the array microphone useless if you want to measure the sound emission from one single source, with other unwanted noise spread throughout $\varphi=-90^\circ$ - 90° . A main lobe is formed when all the microphone signals are added in phase. When the wave falls in with $\varphi=0^\circ$, that occurs. If the wave falls in with $\varphi=90^\circ$ and $d=\lambda$, we also get a summation in phase. The condition for no grating lobes, which determines our upper frequency limit, is:

$$d < \lambda \Rightarrow f_{\max} < c / d \quad (1.2)$$

In our example with $N=8$ and $d=714$ mm, the maximum frequency $f_{\max}=476$ Hz, if we assume that $c=340$ m/s. Note that at this frequency, half a grating lobe is present at $\varphi=90^\circ$ and -90° , so $f_{\max}$ should be slightly lower if we want an array function free of grating lobes.

The lower frequency limit is determined by the total length of the array. If λ is very large, then no difference in phase can be detected by the microphones, and the array microphone becomes omnidirectional, and useless as a tool for directive measurements. In figure 1.3 the contour curve for -10dB is plotted. Here you clearly see the upper and lower frequency limits.

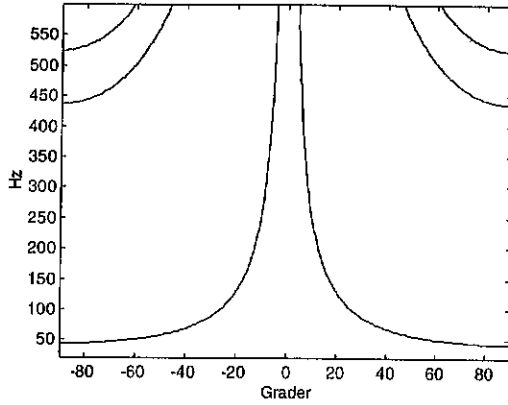


Figure 1.3
-10dB lobe width
 $N=8$, $d=714$ mm.

Note that the grating lobes come in from $\varphi=\pm 90^\circ$.

1.1.2 Linear array with progressive phase

To change the direction of the main lobe, a progressive phase α is introduced. The signal from the summing amplifier then becomes:

$$F = 1 + e^{j(kd \sin(\varphi) + \alpha)} + e^{j2(kd \sin(\varphi) + \alpha)} + \dots + e^{j(N-1)(kd \sin(\varphi) + \alpha)} = \sum_{n=1}^N e^{j(n-1)\beta} \quad (1.3)$$

$$\beta = kd \sin(\varphi) + \alpha$$

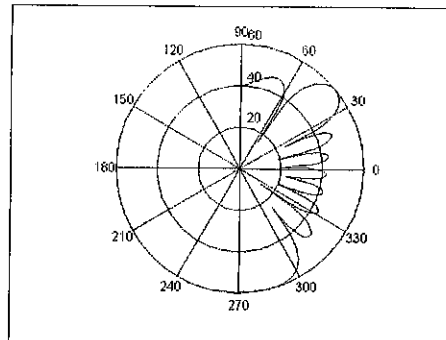
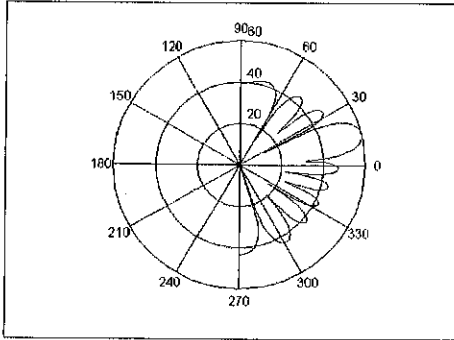


Figure 1.4
 $20\log|F|$
 $\alpha=1$ and
 $\alpha=2,5$.

$20\log|F|$ is plotted in figure 1.4, for $\alpha=1$ and 2,5. You can see a clear dislocation of the main lobe, but at $\alpha=2,5$ the grating lobe is back. You must take in to consideration how far from $\varphi=0^\circ$ the main lobe will be directed. A useful formula for calculating d for a given $f_{\max}$ is given in reference [2]:

$$d = \frac{c}{2f_{\max} \sin(\theta_{\max})} \quad (1.4)$$

$\theta_{\max}$ is the angle outside which you accept grating lobes, and inside which you want to be able to steer the main lobe. If you accept no grating lobes, and want to be able to steer the main lobe from -90° to 90° , then $\theta_{\max}=90^\circ$, and $d=c/(2f_{\max})$.

Another way to approach the problem is to view F as a function of β . Note that $F(\beta + n2\pi) = F(\beta)$, that is β is periodic with $T=2\pi$. The part of $F(\beta)$ which forms $F(\varphi)$ from -90° to 90° , is called the visible region. If $F(\varphi)$ is allowed to contain only one period of $F(\beta)$, then $2kd$ must be equal to 2π , since $\beta = k d \sin(\varphi) + \alpha$. This gives formula (1.5).

$$2kd < 2\pi \Leftrightarrow d < c / 2f \quad (1.5)$$

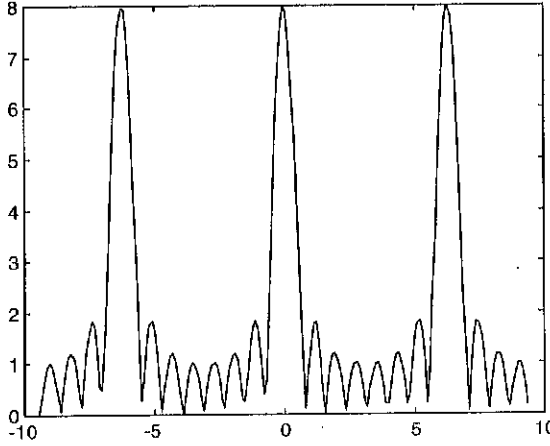


Figure 1.5
 $|F(\beta)|$ for $N=8$
and $d=714$ mm.
 $f=400$ Hz.

In figure 1.5 an example of $F(\beta)$ is plotted. If you are interested in only small deviations of the main lobe from $\varphi=0$, then you can allow yourself to incorporate more than 2π of $F(\beta)$ in the visible region. The extreme case is if the main lobe is very narrow, and you do not want to be able to steer the lobe at all, then you can take 4π of $F(\beta)$. This gives formula (1.6).

$$2kd < 4\pi \Leftrightarrow d < c / f \quad (1.6)$$

This is exactly what we derived in (1.2).

1.2 Weighting

We have seen how we can select d for a simple array microphone, but we still have other parameters to optimize for our system. We can use weighting of the microphone signals to produce some interesting effects, such as the possibility to exchange signal to noise ratio (main lobe to side lobe level) for beam width. If we introduce the weighting coefficients $w(n)$, we get the following expression for F :

$$F = \sum_{n=1}^N w(n) e^{j(n-1)\beta} \quad (1.7)$$

$$\beta = kd \sin(\varphi) + \alpha$$

We choose $\alpha=0$ to make the expressions less cumbersome, but a progressive phase can always be added later. How should we select $w(n)$ to get the desired effects?

1.2.1 The Binomial array

If we choose $w(n)$ as the binomial coefficients we get:

$$F_{arr} = \sum_{n=1}^N \binom{N}{n} e^{j(n-1)\beta} \quad (1.8)$$

$$\beta = kd \sin(\varphi)$$

If we assume $N=8$ and $d=714\text{mm}$, we can plot $|F|$ (figure 1.6).

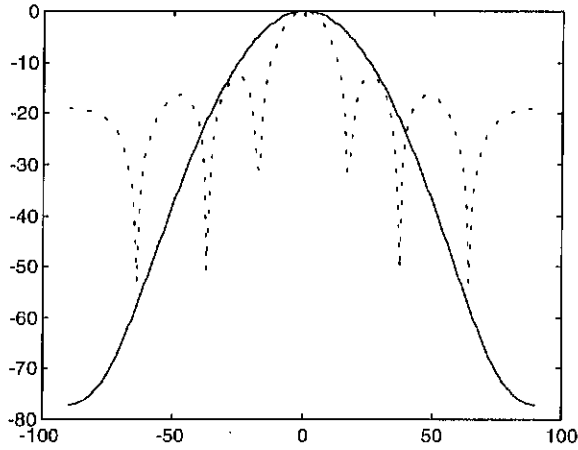


Figure 1.6
Comparison
between the
Binomial array (-),
and the uniform
amplitude array(--).

The binomial array microphone has no side lobes and a rather wide main lobe. If the application involves attenuation of very noisy secondary sources, then the binomial array microphone is a good alternative. However, if the application demands a high spatial resolution, it does not work very well.

1.2.2 Dolph array microphone

The Dolph weighting makes it possible to choose between good SNR, or narrow beam width. By choosing the main lobe to side lobe level, R , the following procedure from [1] calculates the weighting coefficients $w(n)$. It works only for an even number of elements N and for symmetric $w(n)$, which means that $w(n)=w(N-n+1)$.

Write F as:

$$|F| = w(1) \cos(u) + w(2) \cos(3u) + \dots + w(N/2) \cos((2N-1)u)$$

$$u = \frac{1}{2} kd \sin(\varphi)$$

Expand $\cos(mu)$ according to:

$$\cos(u) = T_1(\cos(u)) = \cos(u)$$

$$\cos(2u) = T_2(\cos(u)) = 2\cos^2(u) - 1$$

$$\cos(3u) = T_3(\cos(u)) = 4\cos^3(u) - 3\cos(u)$$

...

where $T_n(z)$ are the Chebychev polynomials.

Choose z_0 :

$$z_0 = \cosh \left[\frac{1}{N-1} \cosh^{-1}(R) \right]$$

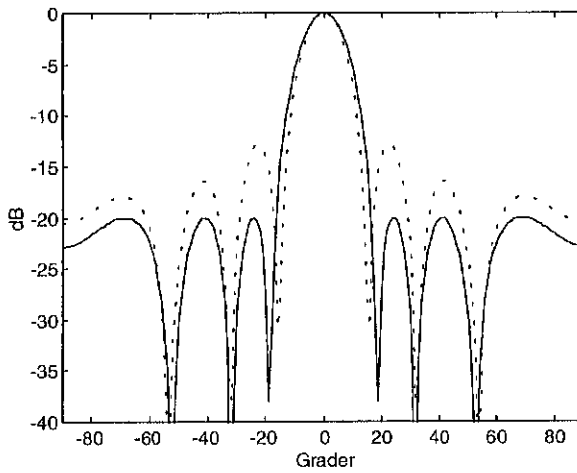
where R is the main lobe to side lobe level (in dB).

Substitute:

$$\cos(u) = \frac{z}{z_0}$$

And finally solve $|F(z)| = T_{N-1}(z)$ by identifying terms, which yields $w(n)$, $n=1,2,\dots,N/2$

If we choose an R which is very large, $w(n)$ will become the binomial coefficients. One could say that the Dolph method lets us design compromises between the uniform amplitude, and the Binomial array microphone. In other words, we can exchange lobe width for SNR, and vice versa.



*Figur 1.7
Comparison
between the Dolph
weighting(-), and
no weighting(--).
 $R=20\text{dB}$*

In figure 1.7 the Dolph solution for $R=20$ dB, and the uniform amplitude solution, are plotted. As usual $N=8$ and $d=714$ mm. Note that the main lobe for the Dolph solution is somewhat wider.

1.3 Microphone spacing

By choosing the microphone spacing, gains in the lower frequency limit can be achieved. Recall that the distance d determines the upper and the total length the lower frequency limit of the array microphone. If the distance between the microphones increases with the distance from the middle of the array, a greater total length can be achieved for a fixed number of microphones, which would affect the lower frequency limit. There is no such thing as a free lunch however, so performance is degraded in other areas.

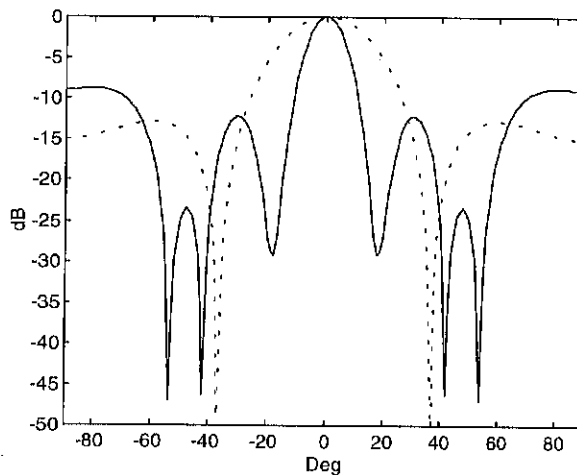
1.3.1 A simple non equidistant array microphone

To be able to study the principal behavior of a non equidistant array, we take the two examples from table 1.1.

Non uniform	Uniformly spaced
-6.7318	-2.5000
-3.8974	-1.7857
-1.7715	-1.0714
-0.3543	-0.3571
0.3543	0.3571
1.7715	1.0714
3.8974	1.7857
6.7318	2.5000

*Table 1.1
Position for the
microphones of
a uniform, and a
non uniform
array in meters.*

The distance between the center microphones is the same for the two examples, but the total length of the non uniform array is larger. If we now calculate $|F|$ for the two examples at 100 Hz (figure 1.8), we can clearly see that the non uniform array main lobe is not as wide as the uniform main lobe, but the main lobe to side lobe level is better for the uniform array. At 400 Hz (figure 1.9), the main lobe situation is the same, but the side lobes are disturbingly high for the non uniform case. The grating lobes occur at the same frequency for both solutions. In other words, we can exchange main lobe to side lobe level for better bandwidth.



*Figure 1.8
 $20\log |F|$ for the
non uniformly (-)
and for the
uniformly (..) spaced array.
100 Hz*

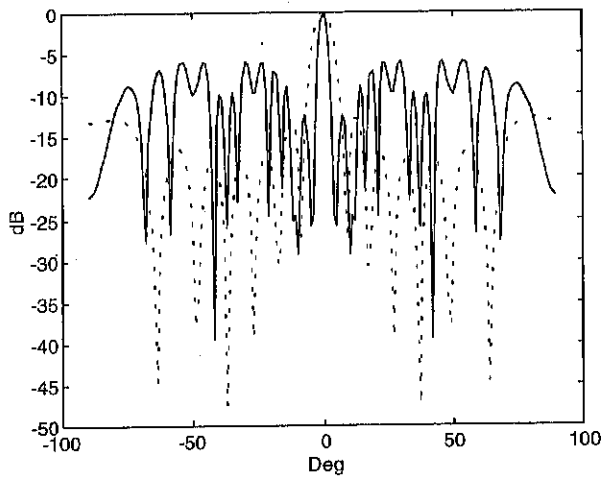


Figure 1.9
 $20\log |F|$ for the
 non uniformly (-)
 and for the
 uniformly (..) spaced array
 400 Hz

It is also possible to optimize the spacing for a given situation, but this requires complicated calculations, such as stochastic region contraction, see [6].

1.4 Constant beam width

In some applications, like measurement of acoustic power, it is desirable to have a main lobe the width of which is constant over the frequency band of interest. If the ratio of total length and wavelength is constant over frequency, so is the beam width. One way of doing this is to have a number of arrays together, possibly sharing the same microphones, and to use them only in a small frequency band corresponding to their length. A much more elegant method is to filter the microphone signals, and, by the help of the filter, create a variable length array.

1.4.1 van der Waal constant beam width array microphone

van der Waal [2] proposes a solution to the constant beam width problem, by filtering all channels with linear phase FIR (Finite Impulse Response) filters, and by using a hybrid between uniform and logarithmic spacing.

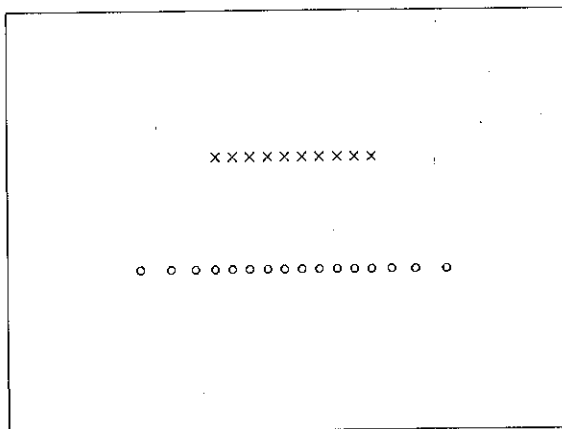


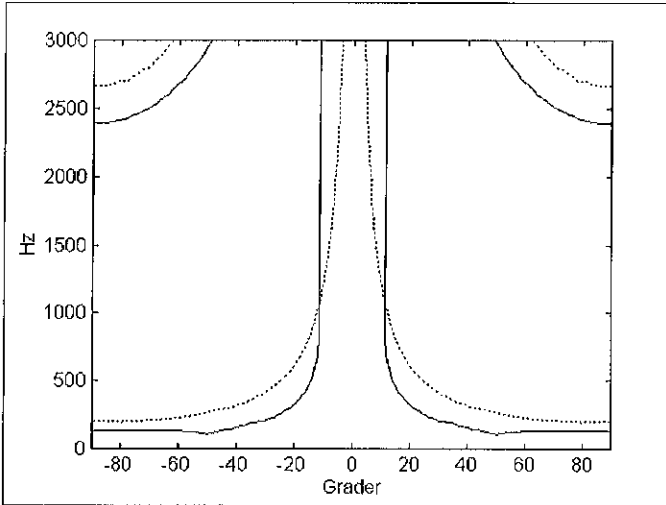
Figure 1.10
 van der Waal
 array with 16
 elements (x),
 and the
 uniform 10
 element array
 which is the
 base (o).

1.4.1.1 van der Waal spacing

The positioning of the microphones is a hybrid between the uniform spacing and the logarithmic spacing. First you design a normal, linear, uniform spacing array microphone, and then you calculate further logarithmically spaced positions to increase the frequency band where the array has a constant beam width. The logarithmically spaced positions are placed according to the formula:

$$\frac{x_{n+1}}{x_n} = \frac{N}{N-2} \quad (1.9)$$

where N is the number of microphones in the uniform part of the array and x_n is the position of the microphone number n , with the center of the array at the origin. The number of microphones placed in the logarithmic part determines the total effective bandwidth extension compared to the uniform part of the array.



*Figure 1.11
10 dB beam
width for
uniform
array(...) and
van der Waal
array(-).*

In figure 1.11, we see the 10 dB beam width of a 10 element uniformly spaced array (dotted), and a 16 element van der Waal array based on the former array. The beam width of the van der Waal array is constant from about 2500 Hz down to 500 Hz, but the main lobe is wider than the uniformly spaced array, even though this one uses only 10 elements. Note that the grating lobes occur at a lower frequency for the van der Waal case.

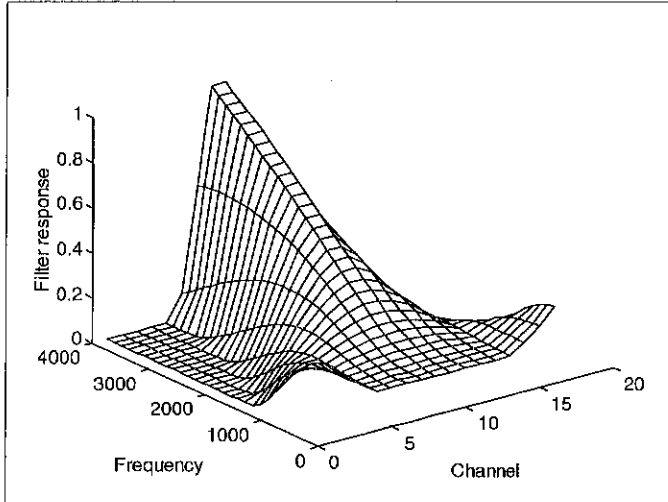
1.4.1.2 van der Waal linear phase FIR filters

The desired frequency response of the FIR filters can be calculated from the microphone positions, and then you have to calculate the impulse response, see 3.1.2. If the microphone positions are x_n , symmetrical around the origin, you can calculate the desired frequency response for the filters from (1.10).

$$T_n(f) = \begin{cases} 0 & |x_n| > \frac{N-1}{2} q\lambda \\ \frac{1}{2} \left[1 + \cos \left(\frac{2\pi x_n}{(N-1)q\lambda} \right) \right] & |x_n| \leq \frac{N-1}{2} q\lambda \end{cases} \quad (1.10)$$

$$q = df_{\max} / c$$

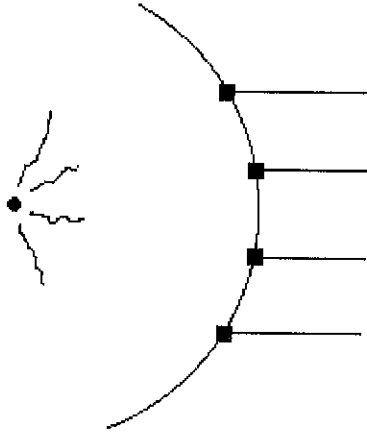
This expression is simply a truncation of the array length, but instead of just letting the filters go directly to zero, a Hanning window is applied. Then you have to correct $T_n(f)$ for the number of active microphones, and for the length each microphone is covering. If we calculate the frequency response for a 16/10 (a total of 16 elements, 10 of which are uniformly placed) van der Waal array, we get the results which can be seen in figure 1.12. For high frequencies, only the center microphones are active, and for low frequencies the uniform part of the array is almost equal in amplitude, with a slight increase for the outer elements.



*Figure 1.12
Filter
frequency
response for a
16/10 van der
Waal array.*

1.5 Focusing the array microphone

All the preceding results assume a plane incident wave. This occurs only if we are very far from the source. If we are close to it, some form of focusing is required. If you want to study a surface of say 1m^2 , and you have a beam width of 20° , then it is impossible to measure from any longer distances. If we assume that the sources of interest are point sources, we have a spherical wave incident on the array microphone. A simple way of focusing the elements is to design the array microphone for the plane wave case, and then reposition them on the spherical wave front. This requires that the distance to the source is known, and that we do not want to steer the main lobe.



*Figure 1.13
Array
microphone
focused by
displacing the
elements.*

If we want to be able to focus at any point in space, we have to use another method. If we introduce a delay in each channel, we can focus on any point in space. The delay times can be calculated, if the sound speed is known. The delays are simply chosen so that all the microphone signals arrive at the same time to our summing amplifier, assuming that we have a spherical propagating wave. Assume that we have a matrix $\mathbf{x}$ with the position of our microphones as columns with x, y, z vectors. Our desired focus point is $\mathbf{p}$.

$$\mathbf{x} = \begin{bmatrix} x_1 & \dots & x_N \\ y_1 & \dots & y_N \\ z_1 & \dots & z_N \end{bmatrix} \quad \mathbf{p} = \begin{bmatrix} x_p \\ y_p \\ z_p \end{bmatrix}$$

The desired delays $\Delta t(n)$ can then be calculated as:

$$\begin{cases} t(n) = \sqrt{(x_n - x_p)^2 + (y_n - y_p)^2 + (z_n - z_p)^2} / c \\ \Delta t(n) = \max_n(t(n)) - t(n) \end{cases} \quad (1.11)$$

Then the following matlab m-file calculates the desired delays for each microphone as a function of microphone position, desired focus point and sound speed:

```

% T=focust(x,p,c)
%
% Calculates the delays for each channel
% for a given focal point p, and sound speed c.
%
% p=[x y z]'
function T=focust(x,p,c);

xp=p(1);
yp=p(2);
zp=p(3);

xdiff(1,:)=x(1,:)-xp;
xdiff(2,:)=x(2,:)-yp;
xdiff(3,:)=x(3,:)-zp;

T=sqrt(xdiff'.^2*ones(3,1))/c;

T=max(T)-T;

```

Note that if we use this form of focusing, no progressive phase need to be introduced to control the main lobe. The direction of the main lobe is controlled by selecting $\mathbf{p}$.

1.6 Effects of phase and amplitude distortion

The microphones used in an array microphone do not have perfectly equal frequency responses, and are not perfectly calibrated together. This will affect the performance of the array microphone. If we assume that this effect can be simulated by introducing a random variation in the amplitude and phase response of the weighting coefficients $w(n)$, we can test how different variations affects $|F|$. We introduce some random variations in $w(n)$, and make a number of realizations to estimate the effect on the array function.

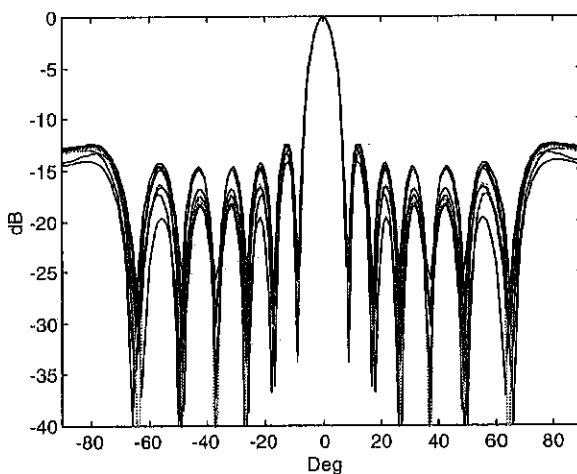


Figure 1.14
 $20\log |F|$ for a
 amplitude
 error of $\pm 10\%$,
 and 8
 realizations.

In figure 1.14 $w(n)=1+b(n)$, where $b(n)$ are normally distributed numbers with variance 0.1. This approximately corresponds to an error of 10%, or in some cases worse, in the calibration. The side lobe levels are affected, but the main lobe is not. This tells us that the amplitude calibration of the microphones is not a critical process for the array performance. If we use some form of weighting, the calibration becomes more important, so we have to see to that the microphones are in a fairly good state of calibration.

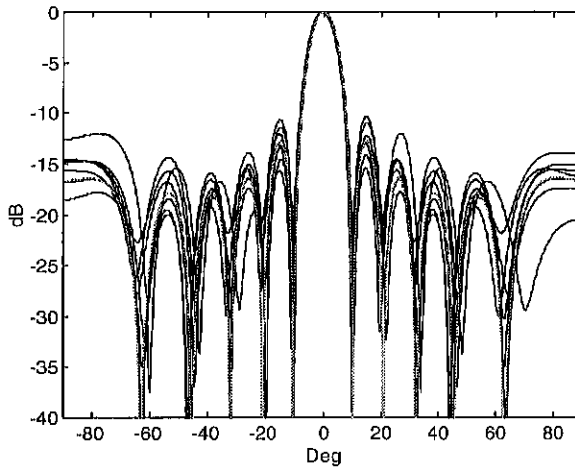


Figure 1.15
 $20\log |F|$ for a
phase error of
 $\pm 10^\circ$, and 8
realizations.

In figure 1.15 $w(n) = e^{j\pi a(n)/180}$, $a(n)$ are normally distributed numbers with variance 10, which corresponds to a phase deviation of about $\pm 10^\circ$. This has a larger effect on the main lobe, and a more random effect on the side lobes, than the amplitude distorted case. In other words, we should use microphones with a good phase match, preferably better than $\pm 10^\circ$.

1.7 Two-dimensional array microphones

The linear array microphones have one common property, they can only suppress secondary noise sources on one axis, see figure 1.16, with the one exception of the end-fire array. The end fire array is the array with the main lobe at $\varphi = 90^\circ$, but it is not further discussed here, since it has a wide main lobe, and it can not be swept. If we want to have a system that can identify sources and suppress secondary sources on a plane, we have to combine one or more linear arrays into a two-dimensional system. Assume that we have a Cartesian coordinate system, and φ is the angle from the unit vector $\mathbf{e}_y$ towards $\mathbf{e}_x$ with $\varphi = 0^\circ$ along $\mathbf{e}_y$. ϕ is the declination, (angle of height above the xy plane). Then the array function becomes a function both of φ and ϕ . If we place a simple linear array along the z-axis, with the center at the origin, the array function $F(\varphi, \phi)$ has the principal behavior plotted in figure 1.16. As can be seen, ϕ has no effect on the array function.

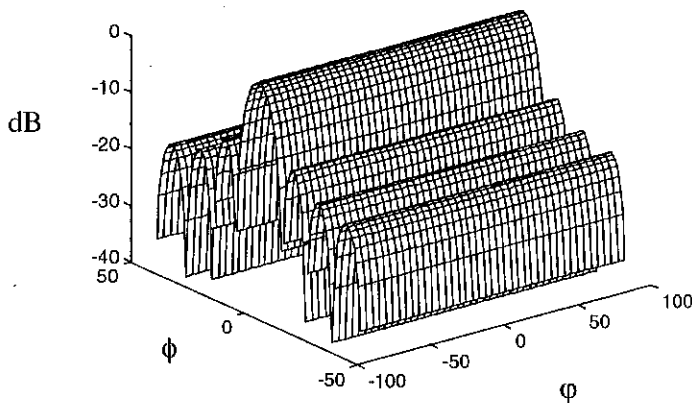


Figure 1.16
 $20\log |F|$ for a
linear array as
a function of φ
and ϕ .

The most direct approach to achieve directivity in two dimensions is to build a matrix of microphones, with $N \times N$ elements in a square formation. If we calculate the directivity of this array microphone, we will notice that the side lobes are high around $\varphi=0$ and $\phi=0$ (figure 1.17)

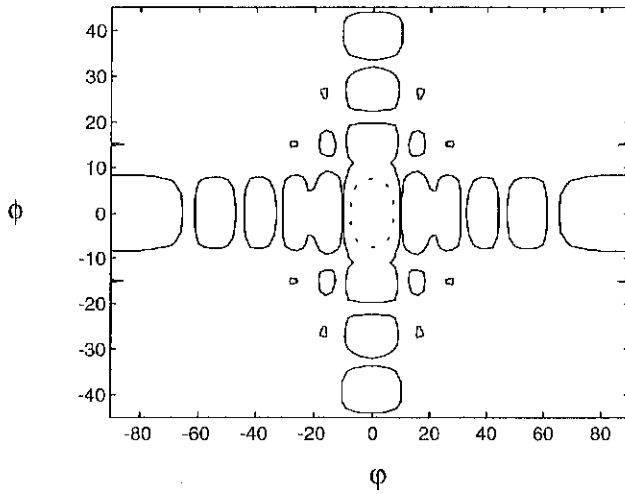


Figure 1.17
The -10(..) and
-20 dB(-)
contour curves
for $20\log|F|$
for a 8×8 array
with $d=8\text{cm}$,
as a function of
 φ and ϕ .
3000 Hz

This form of array microphone uses a lot of microphones, 64 for the 8×8 case, which is a problem. Another solution is to form two linear arrays, and put them perpendicular to each other. The total array signal is then formed from the two separate signals according to (1.12).

$$s_{TOT}(t) = \sqrt{s_1(t)s_2(t)} \quad (1.12)$$

As can be seen in figure 1.18, the main lobe to side lobe level is worse for the cross array, but the beam width is the same as for the matrix case, about $\pm 10^\circ$. If we use some form of weighting we can deal with the side lobe levels, at the cost of a wider main lobe, see figure 1.19. The focusing of the two dimensional array microphone is done in the same manner as for the one dimensional case. Constant beam width can also be achieved by designing the two sub-arrays with a constant beam width approach.

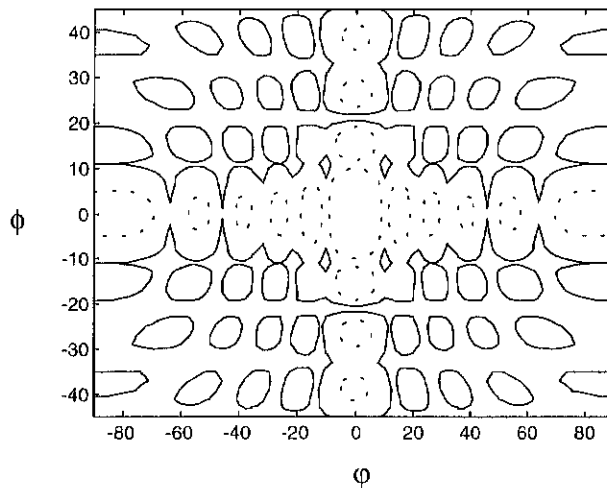


Figure 1.18
The -10(..) and
-20 dB(-)
contour curves
for $20\log|F|$
for a cross 2×8
array with
 $d=8\text{cm}$, as a
function of φ
and ϕ .
3000 Hz

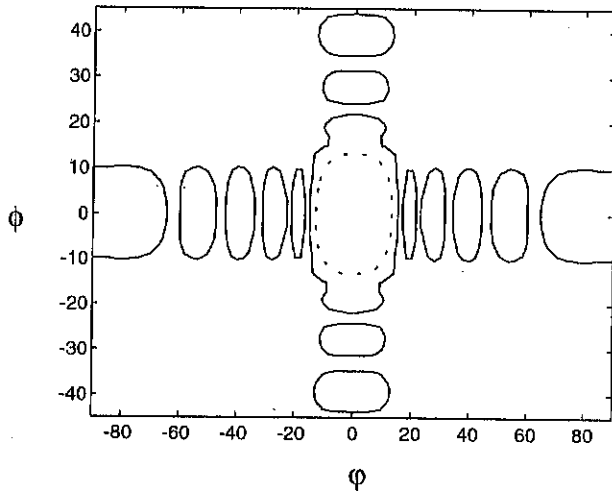


Figure 1.19
The $-10(\dots)$ and
 $-20 \text{ dB}(-)$
contour curves
for $20 \log |F|$
for a Dolph
weighted cross
 2×8 array with
 $d=8 \text{ cm}$, as a
function of ϕ
and Φ .
3000 Hz

2 Implementation and evaluation of a few simple array microphones

2.1 Initial measurements

As an introduction to the diploma work, and to get familiar with the available equipment, I made some simple measurements.

2.1.1 The two microphone array

The measurement was made in the hemi anechoic chamber at SP. To reduce the effect of the floor reflex, three sheets of 70mm mineral wool were placed on the floor. The microphones were mounted on an aluminum profile about 1,5m above the floor.

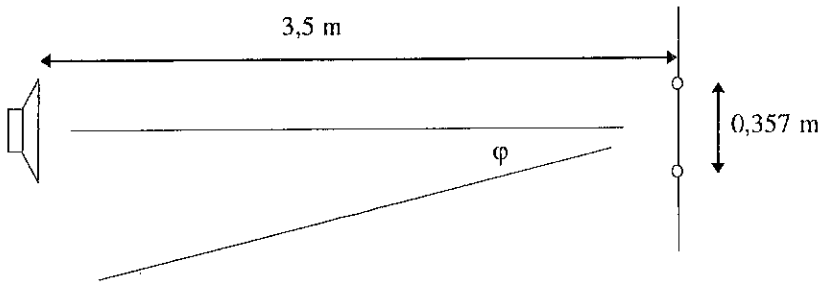


Figure 2.1
Measurement set up.

The data was collected with an DAT tape recorder (SP 502186), and sampled to a computer.

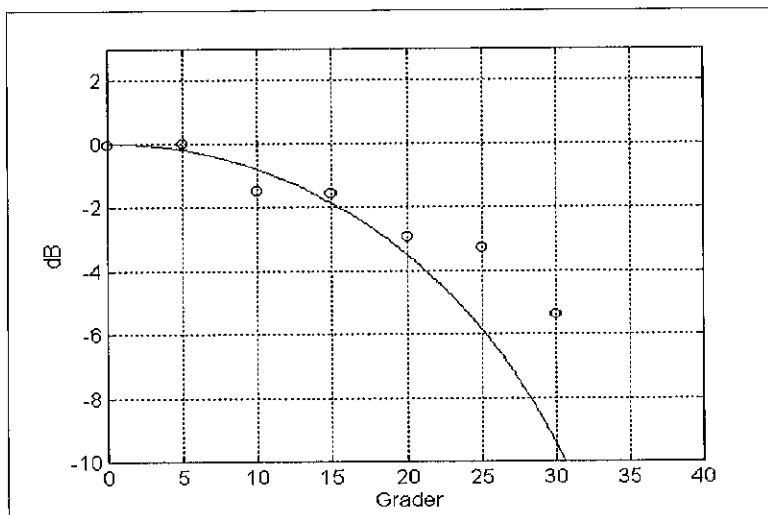
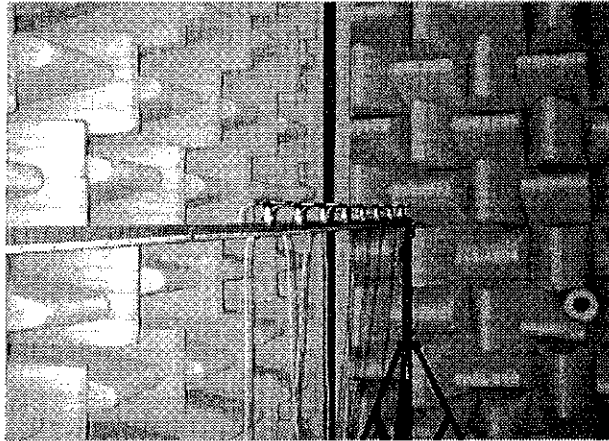


Figure 2.2
Theoretical and
measured $20\log|F|$
 $N=2$, $d=0.357m$
 $f=750$ Hz

The result shows that a directivity pattern can be realized with relatively simple means.

2.2 An eight element linear array with uniform amplitude and spacing

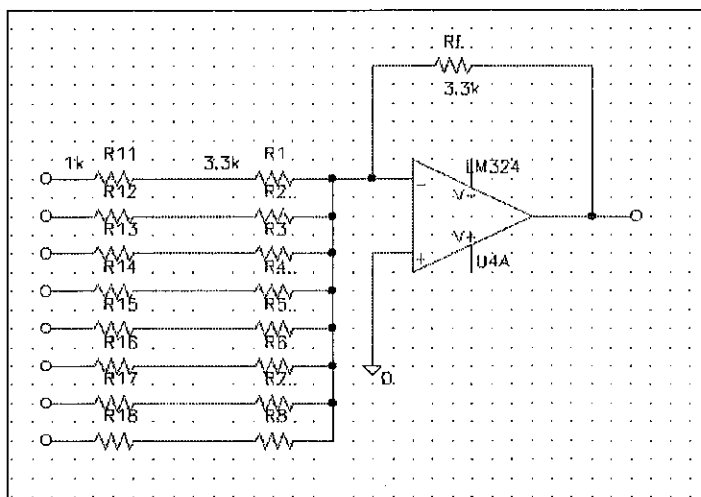
The second array microphone to be constructed was an eight microphone linear array, with uniform amplitude and spacing. It was originally intended to record the eight channels to a DAT-system, but no practical method to transport the data from DAT to a computer based analyze system without the loss of phase information was available. To reduce the number of channels to be recorded, an analog summing amplifier was constructed.



*Figure 2.3
The eight
microphone
array placed in
the hemi
anechoic
chamber at SP.*

2.2.1 Analog summing amplifiers

A Bruel&Kjaer channel selector type 5619 with optional summing function installed, was used to sum up the signals. During calibration it was noted that the channels had somewhat different gains, so a second summing amplifier was constructed, with the operational amplifier TL072 as the active element. Variable resistors were incorporated to allow a good calibration of each channel with respect to the others, se figure 2.4.



*Figure 2.4
Analog summing
amplifier based
on the
operational
amplifier TL072.
(not LM324)*

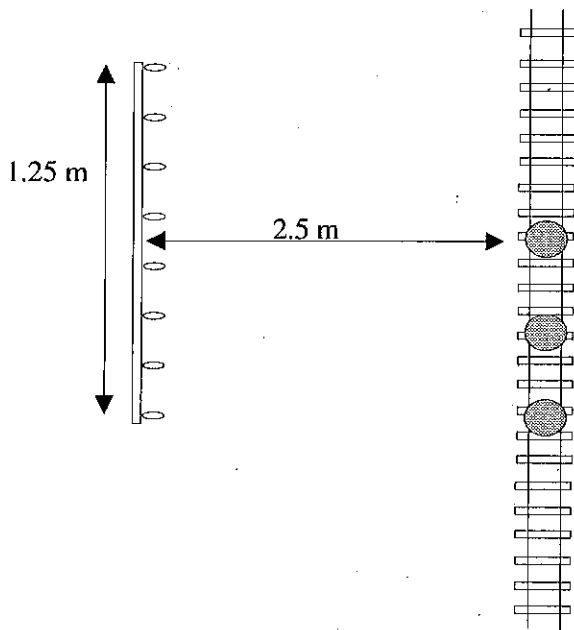
2.2.2 Measurement in hemi-anechoic chamber

The eight element uniform spacing and amplitude array microphone with $d=17,9$ cm was placed in a hemi-anechoic chamber at SP. A nozzle with four crossing streams of pressurized air was used as the source. The microphones were Larsson&Davies 2541, with four B&K 2619 preamplifiers and four Norsonic 2200 preamplifiers. First the microphones were connected to a B&K Front End 2966, and calibrated through the 5619 summing amplifier, and the results are shown in table 2.1.

Microphone no	SPL dB	Table 2.1
1	94 (ref)	Total level of
2	94,9	each channel
3	95,1	from
4	96,5	microphone
5	95,6	through the
6	96,4	B&K 2966 to
7	95,9	the analyzer.
8	95,4	

As can be seen, the result implied that the channels where not so well matched, but the measurement was carried out nevertheless.

The second measurement was made with the TL072 summing amplifier, calibrated to ± 0.2 dB.



*Figure 2.5
Measurement
setup. The
distance
between the
source
positions is
20cm.*

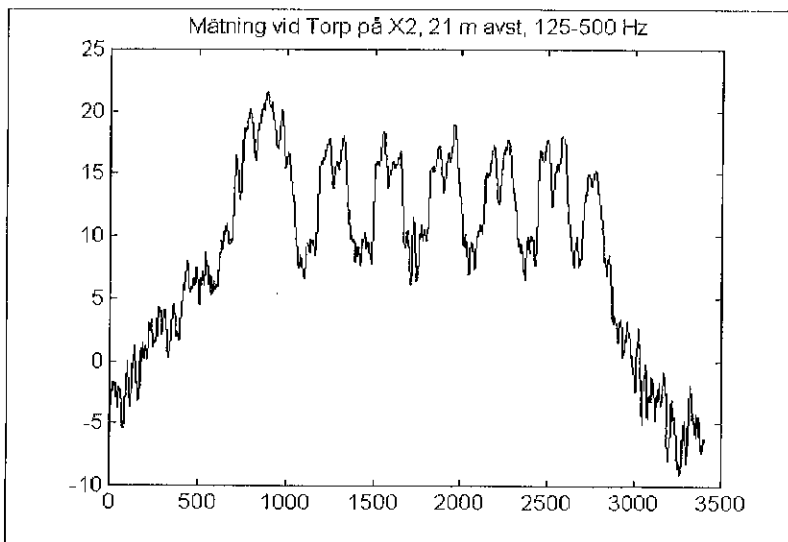
The source to receiver distance was 2.5 m. The source was moved between the measurements with 20 cm at the time, from a center position to a final position 2 m from the starting position. At each position the L_{eq} -value with $T=10$ s was measured in third-octave bands. The resulting diagrams can be found in **annex A**. Note that the lobe width is approximately equal for all frequencies, this is due to the fact that the array is focused for plane waves. As can be seen, the B&K summing amplifier measurement have somewhat higher side lobe levels, but not as much as one would expect from the

calibration results. No distinct nulls can be seen in the measured data, this is due to that we measure in frequency bands, not for a single frequency which is the case for the theoretical values.

To get a clear view of the effects of focusing the array, a displaced configuration was implemented, where the microphones were placed along the spherical wave front from the center position. A new set of measurements were made, and the resulting diagrams can be found in **annex B**. Now, a clear sharpening of the main lobe can be observed for higher frequencies. This tells us that the focusing is important if we want a high spatial resolution, but it also informs us that it might be useful to place the microphones at an optimum distance, in order to get a constant beam width over frequency, which we almost had during the first two measurements.

2.2.3 Measurement of passing X2 train at Torp

An linear, uniformly spaced array with $N=8$ and $d=714\text{mm}$ was positioned 21 m from the railway track at Torp, about 60 km to the north of Gothenburg. The microphones used were Larsson&Davies 2145, and these were connected through four Norsonic 2200 and four Bruel&Kjaer preamplifiers to a B&K Front End 2966. The microphone signals were summed up by the TL072 summing amplifier described in 2.2.1. The array microphones distance to the center of the rail was 21 m. One passage of a X2-train was recorded on DAT tape, and later analyzed with FFT. The beam width at this distance is about 5m, and the wheels are placed 2.9 m apart on a X2 train, so one can not expect to see every single wheel, but the boggies can be seen as distinct tops in figure 2.6. Note that no calibration was made, and therefore the values on the y-axis are interesting in a relative manner only.



*Figure 2.6
Measurement
of a X2 train
bypass at Torp,
125-500 Hz.
x-axis is time
in unspecified
units.*



*Figure 2.7
The array
microphone at
Torp.*

3 Design and evaluation of a complete array microphone system

3.1 The measurement system

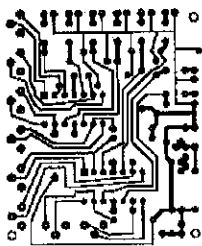
No system which could record more than two channels simultaneously into a computer was available at the beginning of the diploma work, and therefore one had to be constructed. The specification was set at 16-32 channels at a minimum sampling frequency of 10kHz per channel. A 12-bit A/D board was considered sufficient. Finally a 12 bit, 16 channel board was selected, but two separate boards were purchased, so that they could be synchronized and a total of 32 channels would be available.

The boards were installed in a PC, and programmed to wait for a trigger signal, then simply collect data into the RAM memory for a preset number of samples. After the sampling is ready, the data is saved to disk. The data can then be loaded into MATLAB for analysis.

3.1.1 Microphones, anti-alias filters and pre-amplifiers

A few different types of electret microphones were evaluated, both omnidirectional and unidirectional ones. To eliminate sound from behind the array microphone, and because its good and simple electrical behavior, a UEC-14 microphone was selected.

The preamplifiers were constructed around the TL074 operational amplifier, used in the non-inverting amplifier mode, to ensure high input impedance. Two TL074 capsules were placed on one board to allow eight channels. The input impedance is $1\text{M}\Omega$ parallel to $0.1\mu\text{F}$. The board layout is presented in figure 3.1.



*Figure 3.1
Board layout
for the
preamplifiers.*

The anti-alias filters were constructed around the Maxim 293 eight order elliptic switched capacitor filter circuit. Pre filtering and high input impedance were ensured by a TL072 amplifier on each channel. The Max293's internal operational amplifier was used for post filtering. A CMOS 555 circuit provided the clock signal, which was trimmed to 400kHz, for a cutoff frequency of 4kHz.

To test the filter performance, the frequency response was measured with MLSSA. As can be seen in figure 3.2, the signal to noise ratio is about 65dB, and the attenuation from 4kHz to 5Khz about 40dB, which makes it possible to set the sample frequency to 10kHz without risk for aliasing errors. The dynamic range of the filter, 65dB, is about what we get if we use a 12-bit A/D converter.

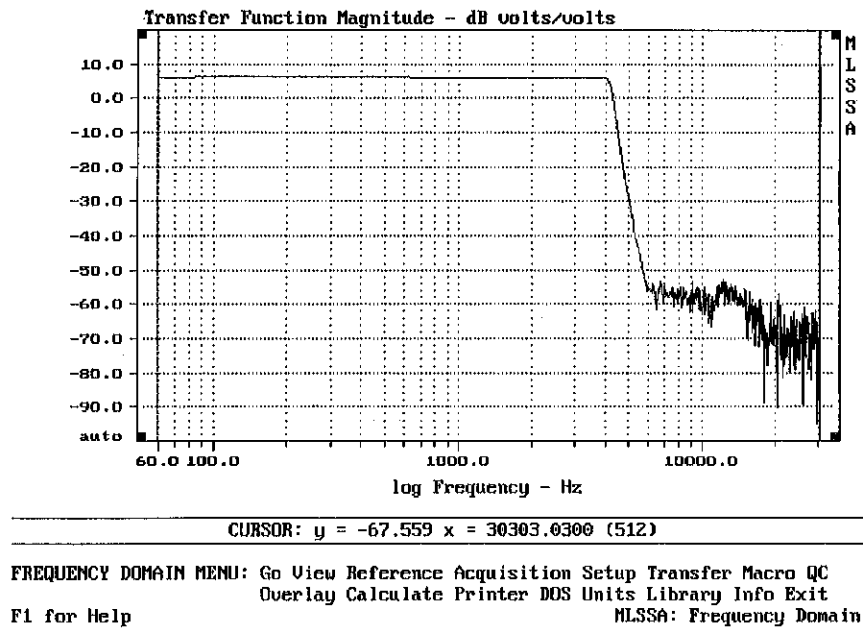


Figure 3.2
Measured
frequency
response of
the anti alias
filter.

The board layout is shown in figure 3.3. To get low crosstalk between the channels, each channel was separated from the others, and no components were shared between channels. Each channel has one TL072 operational amplifier and one MAX293 filter IC, with their own separate decoupling capacitors.

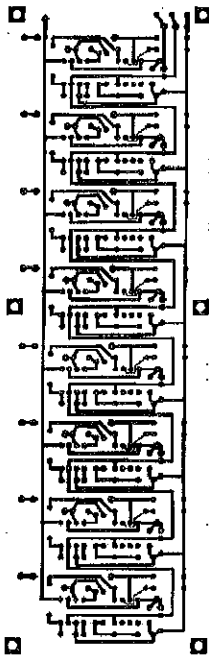


Figure 3.3
Board layout
for eight
channels of the
anti alias filter.

3.1.2 Analysis programs in MATLAB code

After the data is stored, some form of analysis is necessary. Since all the microphone signals are recorded, it is possible to use different kinds of weighting, and to sweep the main lobe, to analyze the data. The most simple form of analysis, is to form a main lobe, and to use it through all the time data. A more complicated form is to change the main lobe direction as a function of time. Consider the situation of a vehicle passing the array,

and we want to analyze the sound emissions. If the array microphone is a linear array, we can analyze parts of the vehicle in one direction, but we can not discriminate between two sources if they are perpendicular to this direction. If we use a two dimensional array, we can analyze everything in a plane in front of our array, but we also need to control more parameters in our analysis programs. The basic parts of our analysis are:

Correction for non simultaneous sampling

First of all, we need to get the samples corrected for the sweeping sampling process. The sampling board uses a multiplexer to switch between the channels and to sample each channel in turn. This means that the first channel gets sampled at t , then the second get sampled at $t+\Delta t$, the third at $t+2\Delta t$, and so on. $\Delta t=1/(F_s K)$ where K is the number of channels. If we know that the signal is band limited to $F_s/2$ Hz, then we know from the sampling theorem that the signal x is known for all t , if we know it for all $x(nT)$, where $T=1/F_s$ and $n=\dots,-1,-2,0,1,2,\dots$. We can in other words reconstruct the samples $x(\Delta t+nT)$ to another set of samples $x(nT)$. Since the A/D board in question has 16 channels, we interpolate the signal to increase the sampling rate by a factor 16. The interpolation fills the spare positions with zeros, and then lowpass filter the signal. Then we sample down to the original rate by selecting the correct samples from the interpolated version. This method uses 16 times the memory of the signal one wishes to correct, so a hardware sample and hold would reduce the processing time. The following MATLAB m-file does the correction:

```
% L=kansep(p)
%
% This m-file takes a sampel vector p, and separates
% the 16 channels. It also corrects for incoherent
% sampling by resampling with a higher rate, and then
% selecting the correct samples. Channel 16 is presumed
% to be sampled first.
%
% The output is a matrix L, with each channel as a
% row vector.

function L=kansep(p);

M=length(p);

for n=1:16
    temp=resample(p(n:16:M),16,1);
    L(n,:)=temp((17-n):16:length(temp));
end
```

Another method to achieve the same result is to use a filter with the (ideal) frequency response from (3.1) to filter each channel.

$$H(\omega) = e^{-j\omega n_0} \quad (3.1)$$

This filter will delay the signal with n_0 samples, even if n_0 is not an integer, and by this method we can delay all channels so they appear to have been sampled at the same time as the last channel to be sampled. The filter can be implemented as a FIR filter if we calculate the impulse response $h(n)$, truncate to finite length and apply a window function. The calculation of $h(n)$ can be done as:

$$h(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(\omega) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-j\omega n_0} e^{j\omega n} d\omega = \frac{1}{j2\pi(n-n_0)} \left[e^{j\omega(n-n_0)} \right]_{-\pi}^{\pi} =$$

$$= \frac{\sin(\pi(n-n_0))}{\pi(n-n_0)}$$

This method is more time efficient than the other one presented, but the first method is perhaps more straight forward. Note also that since the impulse response has to be truncated to finite length, the frequency response is no longer ideal, and we introduce errors, especially close to the Nyquist frequency ($F_s/2$). Note also that long filters are accurate, but time consuming. The following MATLAB file calculates the desired filter impulse response to correct for the non simultaneous sampling:

```
% This function calculates the necessary filters to correct for
% non simultaneous sampling in the CIO-DAS 16/330 Board.
%
% hout=delfilt();
%
% hout is a matrix with 16 rows corresponding to the channels,
% each containing a 32 length FIR filter.
%
function hout=delfilt();
n=-15:16;
h=sinc(n).*hanning(32);
hout(1,:)=h;
for ko=2:16
    k=ko-1;
    h=1./(pi.*(n-k/16)).*sin(pi*(n-k/16));
    h=h.*hanning(32);
    hout(ko,:)=h;
end
```

Bandpass filtering

After having corrected the samples to the same time base, a bandpass filtering is a good idea. Then we get rid of any DC-offset from the anti alias filters, or from the A/D board, and we can ensure that no frequencies higher than the array microphone upper frequency limit slips through our analysis. Note that this filter does not have to be of linear phase, because we already have the signals recorded, so we can filter both in the backward and forward direction. This means that no phase change will affect the signal no matter what filter we use. (MATLAB command `filtfilt`)

Constant beam width

If we want to use a constant beam width solution, then we would have to filter each channel individually. Since we do not work in real time, we do not have to use a linear phase filter, but since the design algorithms are relatively easy for FIR linear phase filters, we will use this approach. The filter coefficients for a linear phase FIR filter can be found as the solution to a matrix equation $\mathbf{A}\mathbf{h}=\mathbf{H}$, see page 342 in [4]. The following MATLAB m-file calculates $h(n)$ for a give frequency response $H(k)$.

```
function h=linfir(H);
N=length(H);
M=2*N;
k=1:N;
wk=2*pi*(k-1)/M;
for n=1:N
    for k=1:N
        A(k,n)=2*cos(wk(k)*((M-1)/2-(n-1)));
    end
end
ht=A\H';
h=[ht' flipr(ht)'];
```

Focusing

To focus the main lobe on a point in space, we have to calculate the desired delays for each channel, se 1.5, and then impose them as either a padding with a corresponding number of zeros in the beginning of the signal, or as a filter with $h(n)=[0 \ 0 \ \dots \ 0 \ 1 \ 0 \ \dots \ 0]$. If we use some form of FIR filtering, we can combine the filters into a single filter.

There is one problem though, if we are using the sampling frequency 10kHz, we will have a minimum delay unit of 0.1ms. This means that smaller delays cannot be represented. If we have a uniform spacing array microphone, with $N=8$, $d=8\text{cm}$ and focusing on a point 3m in front of the array, the desired delays would be 0, 0.0188, 0.0313, 0.0376, 0.0376, 0.0313, 0.0188, 0 ms, assuming $c=340\text{m/s}$. This means that no delay, and thus no focusing, can be implemented. The solution to this is once again interpolation, or filtering in the same manner as described in correction for non simultaneous sampling. If we interpolate to 10 times the sampling rate, we make the minimum delay unit 10 times smaller, and can focus the array without significant problems. The interpolation uses memory and takes time, so for a optimum solution, one should not use too high an order of interpolation, or to long filters if that option is used.

3.2 Initial measurements with simplified setup

To test the measurement system built, a simple array microphone with $N=8$ and $d=8\text{cm}$ was constructed. The array microphone was placed in the hemi anechoic chamber at SP, and a speaker was placed 3.3m in front of the array.

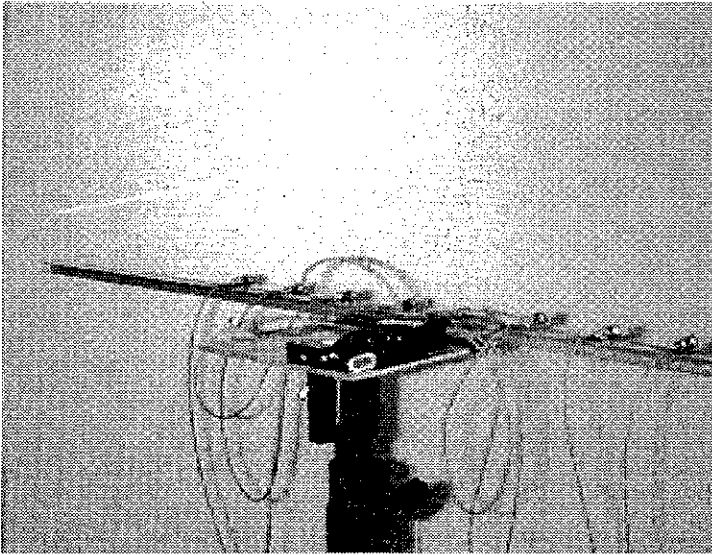


Figure 3.4
The $N=8$,
 $d=8\text{cm}$ array
test
configuration.

A white noise was emitted from the speaker, and the 8 microphone signals were sampled directly to the computer RAM memory. During analysis, the array microphone was focused on the speaker, and then refocused on positions to the right of the speaker with an interval of 20 cm. With this technique we can get a look at our main lobe by sweeping it instead of moving the source. The measured data was filtered with a bandpass filter with passband 3500-4000 Hz, and then the power of the signal was evaluated with a FFT method. The resulting lobe can be found in figure 3.5, together with the theoretical curve calculated in MATLAB for 3750 Hz. Note that the theoretical function is not focused.

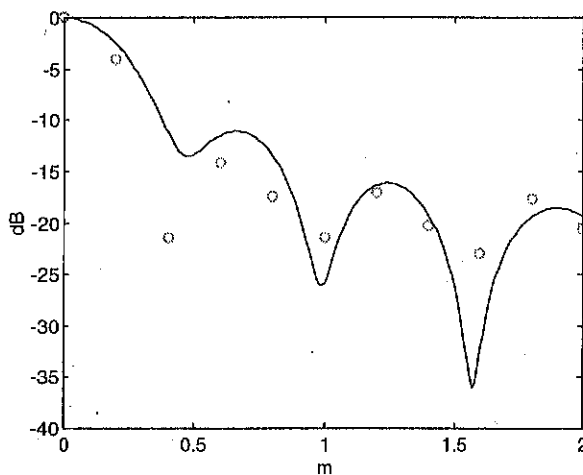
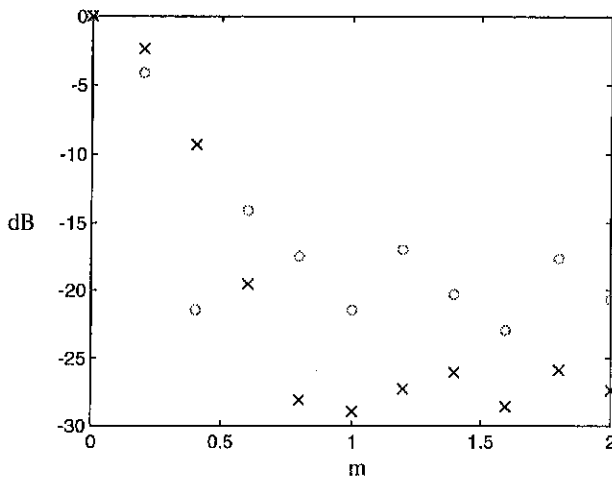


Figure 3.5
Measured (o)
and
theoretical(-)
array function
3500-4000 Hz

To see the effects of weighting, a Dolph weighting was introduced (see 3.6) with $R=30$ dB. The main lobe to side lobe level has increased, but not quite to 30 dB. This is due to crosstalk in the cables, which has been measured to about 20 dB. Note also the widening of the main lobe when we use Dolph weighting.



*Figure 3.6
Measured
directivity,
Dolph (x) and
uniform(o)
weighting
3500-4000 Hz*

The test proved that the system can successfully handle the task of forming a main lobe for frequencies up to 4kHz.

3.3 Comparison between the array microphone and a parabola

Another method to achieve a directivity in acoustic measurements is to use a parabolic reflector. This method uses only one microphone, placed at the focus of the parabolic reflector. The main limitation of the parabola compared to an array, is that it has a stationary main lobe, it can not be swept during or after a measurement. The parabola does not have an upper frequency limit like the array, but it does have a lower frequency limit. When the incident wave decreases in frequency, sooner or later no reflection will be present because the wavelength λ becomes big in comparison to the diameter of the parabola, and the parabola becomes omnidirectional.

To compare the performance of a parabola with that of the array, a series of measurements were made. A parabola with 2 m diameter was set up on a lawn at SP, and a speaker was placed 10 m from the focus and moved in a circle around his point, in steps corresponding to 1° . The transfer function was measured with MLSSA for different angles of incidence. Finally the directivity for one third octave bands was calculated from the measured transfer function. The parabola is not immediately comparable with the linear array because the parabola is two dimensional, but a two dimensional array can be constructed from two or more linear arrays, see 1.7.

To compete with the parabolas high frequency characteristics, the array function of a uniform linear array with 16 elements were calculated, with $d=8$ cm, which corresponds to an upper frequency limit of 4kHz. Higher frequencies are not considered here, since the array system constructed has a maximum frequency limit of 4kHz due to sampling restrictions.

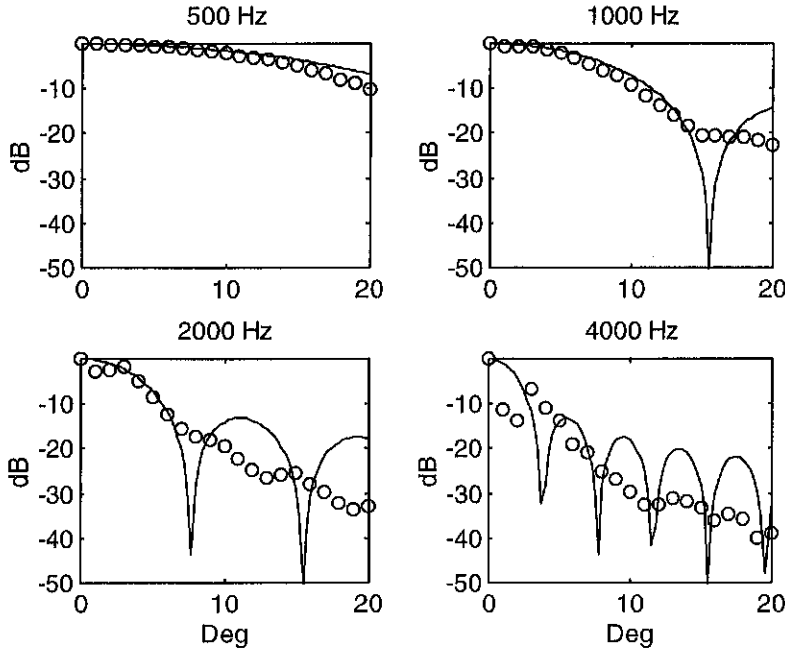


Figure 3.7
Parabola
measurements (o)
compared to array
calculations (-).
N=16, d=8cm
Third octave
values for the
parabola, and
single frequency
values for the
array.

As can be seen in figure 3.7, the parabola has a very sharp characteristic for 4kHz, but it also have a side lobe up to -5 dB. For lower frequencies, the beam width is fairly equal, with a slight tendency to a side lobe for the parabola at 2kHz.

For the lower frequency region, a similar solution for 1000 Hz and downwards with $d=28.3$ cm and $N=16$, which gives a total length of about 4.2 m, is compared to the parabola, se figure 3.8.

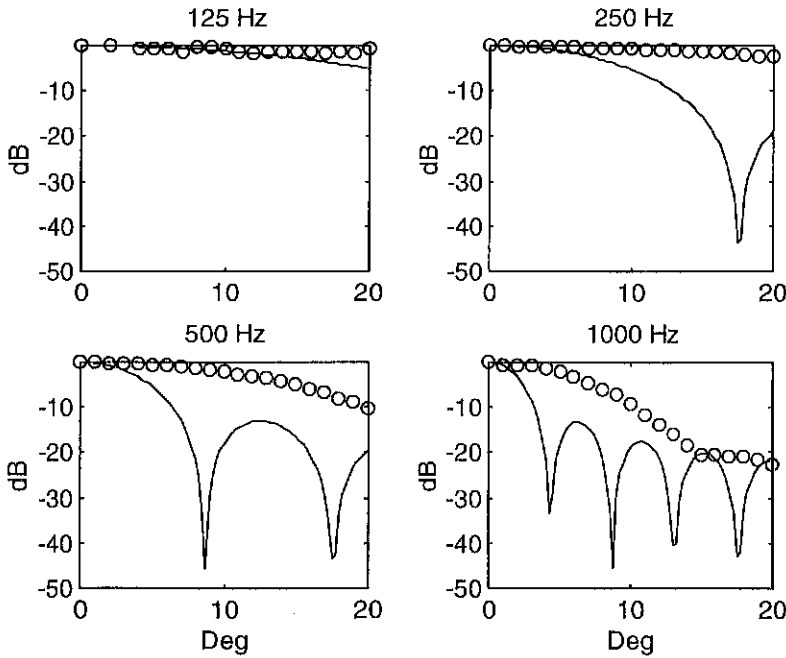


Figure 3.8
Parabola
measurements (o)
compared to array
calculations (-).
N=16, d=28.3cm
Third octave
values for the
parabola, and
single frequency
values for the
array.

Finally a constant beam width solution of van der Waal type, with 10 uniformly spaced and a total of 16 microphones, was compared to the parabola. The upper frequency limit was set to 500 Hz, and the angle $\theta_{\max}$ was set to 45° , se 1.1.2 formula (1.4). The total length of the array is 8.4 m, a rather cumbersome size if one wants to construct this

solution in two dimensions. The array functions and the directivity of the parabola is plotted in figure 3.9.

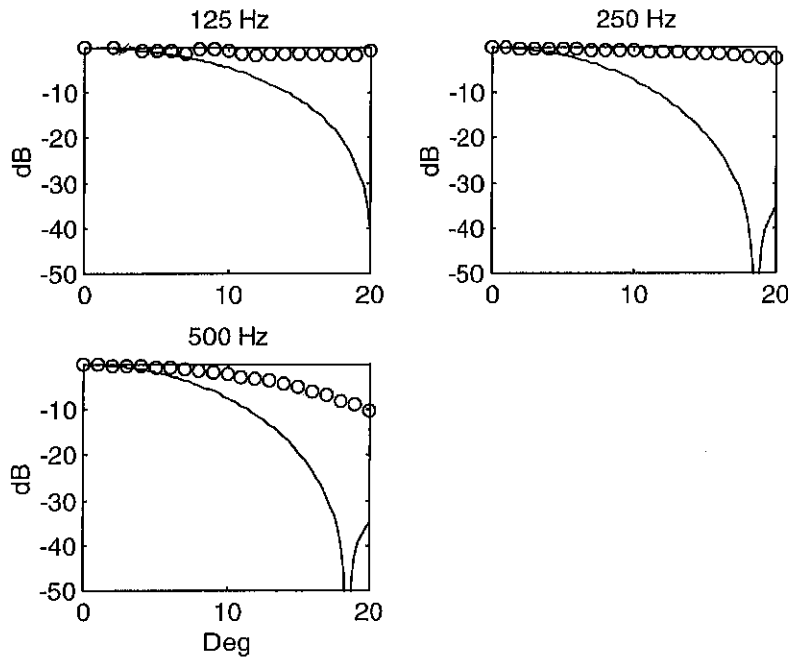


Figure 3.9
Parabola
measurements (o)
compared to array
calculations (-).
van der Waal
10/16 array
Third octave
values for the
parabola, and
single frequency
values for the
array.

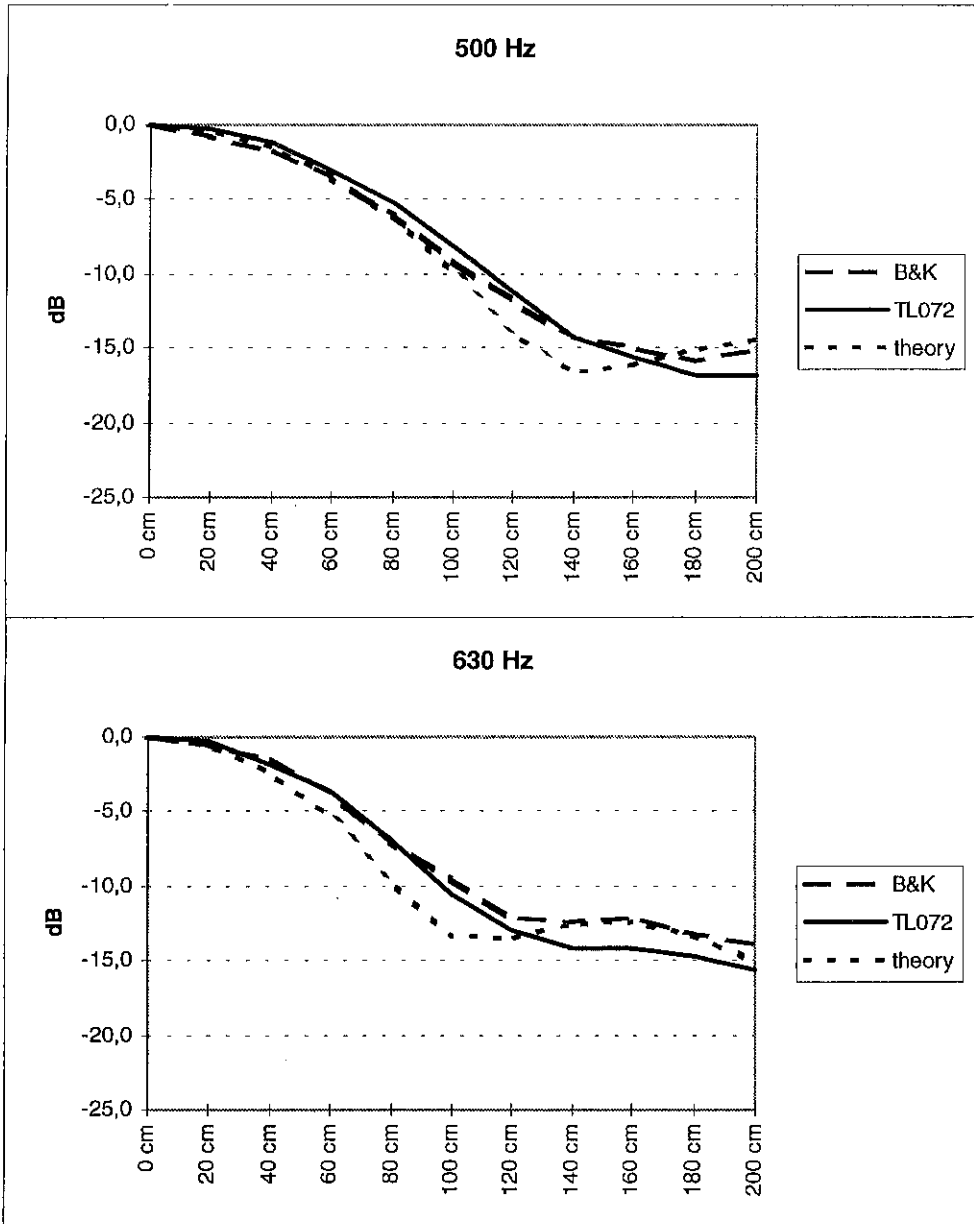
The array system developed can easily beat the performance of the parabola, in terms of lobe width, for low frequencies. The parabola has however other advantages, such as low cost, easy maintenance and little or no post processing time.

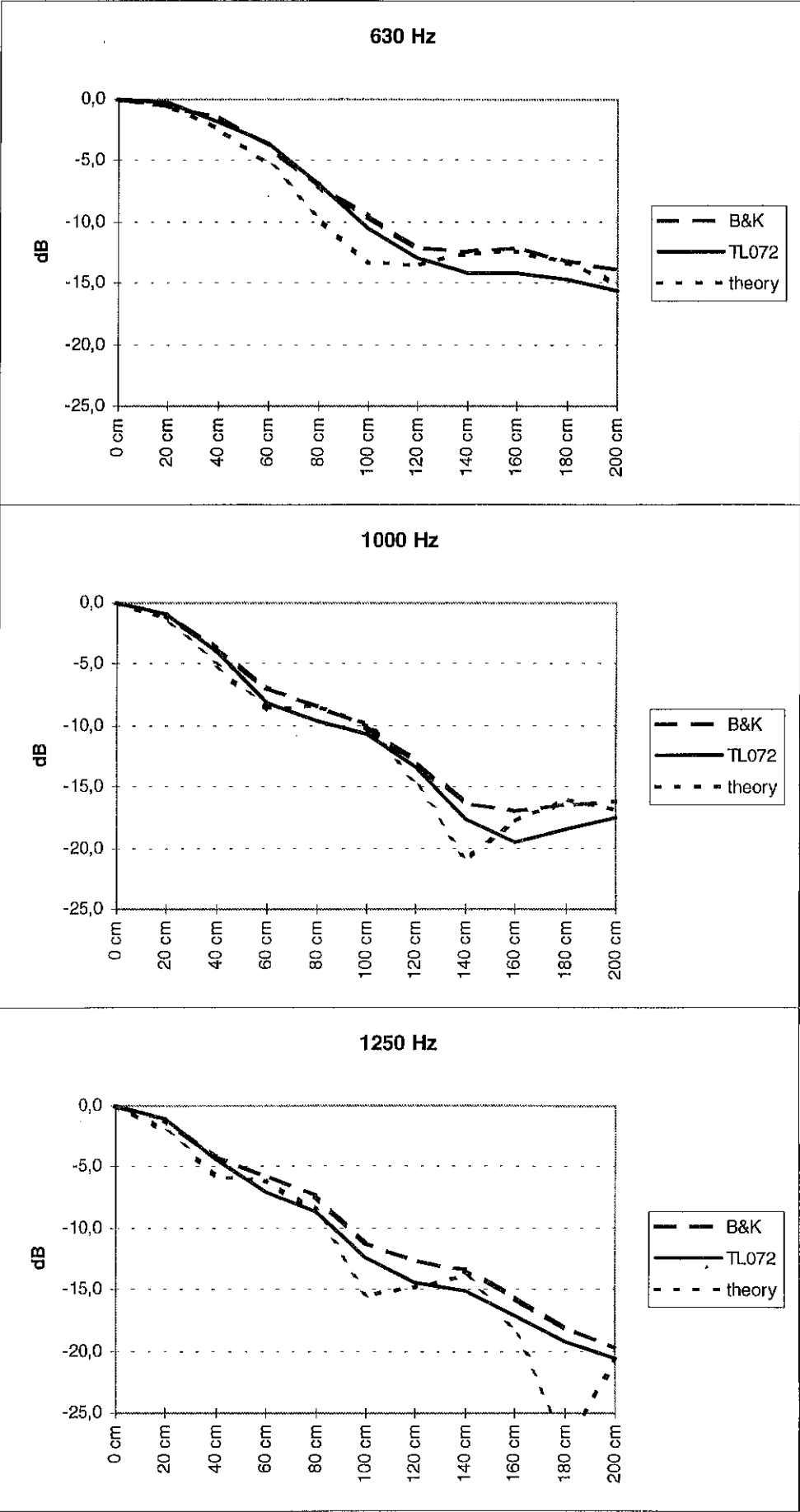
References

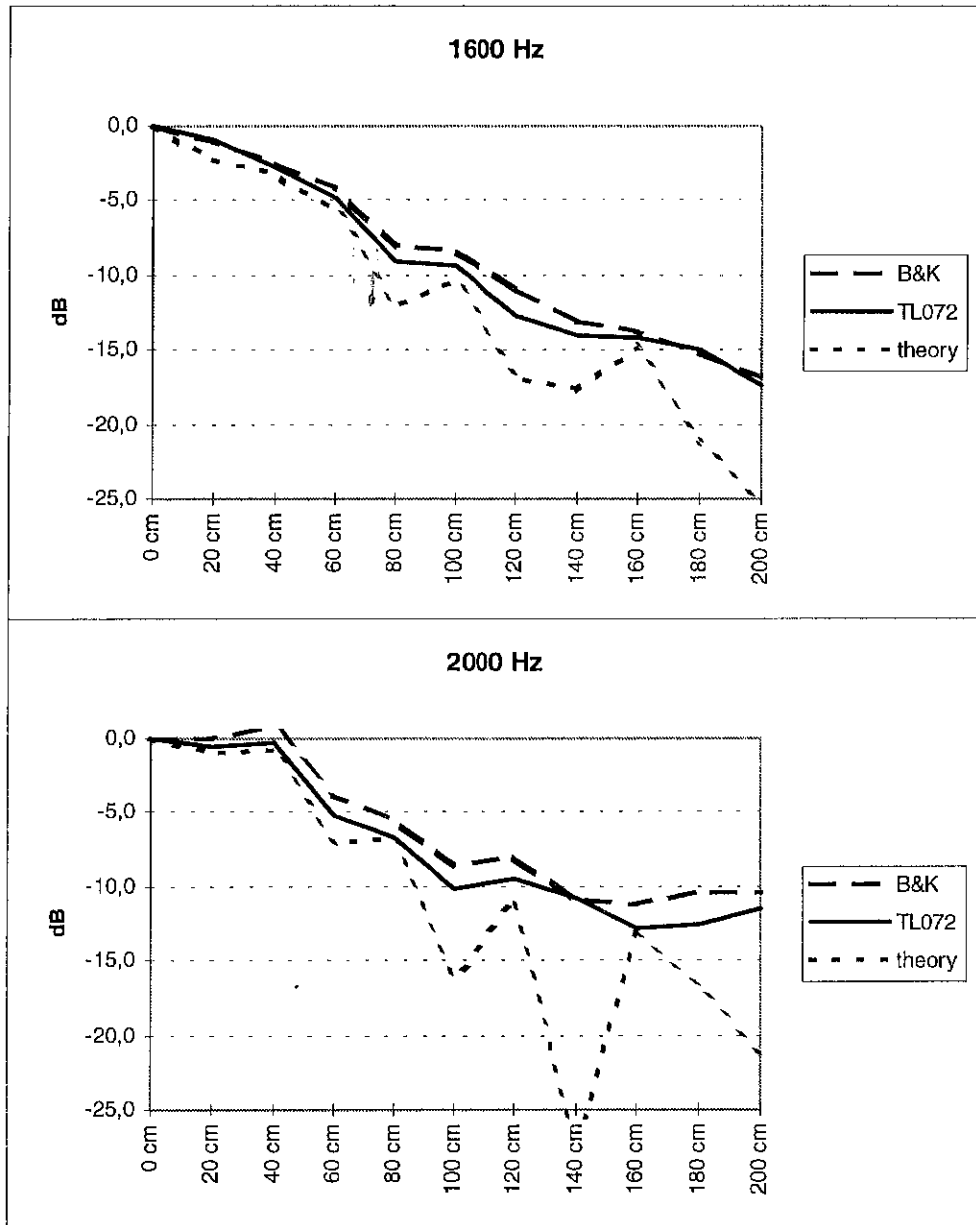
- [1] Constantine A. Balanis
Antenna Theory: Analysis and Design
1982
- [2] Menno van der Waal, Evert W. Start, Diemeir de Vries
Design of Logarithmically Spaced Constant-Directivity Transducer Arrays
AES, Vol. 44, No. 6, 1996 June
- [3] Carsten Sydow
Broadband beamforming for a microphone array
J. Acoust. Soc. Am., 96(2), Pt .1, August 1994
- [4] John G. Proakis, Dimitris G. Manolakis
Digital Signal Processing: Principles, Algorithms, and Applications
Second Edition, Macmillan Publishing Company, 1992
- [5] A. A. (Lois) Beex, Victor E. DeBrunner
Restricted Geometry Acoustic Arrays for Highly Directional Patterns
Applied Acoustics, vol 33, No 1, 1991
- [6] Victor M. Alvarado, Harvey F. Silverman
Experimental Results Showing the Effects of Optimal Spacing Between
Elements of a Linear Microphone Array
ICASSP 90. 1990 International Conference on Acoustics, Speech and
Signal Processing (Cat. No.90CH2847-2), p. 837-40 vol.2
- [7] Daniela Profir, Emil Sunvisson
Riktmikrofoner - en översikt
Inst. för farkostteknik, Kungl. tekniska högskolan, Stockholm 1995

Annex A, Measurement in hemi-anechoic chamber

Measurements on a linear uniformly spaced array with eight microphones, se 2.2.2

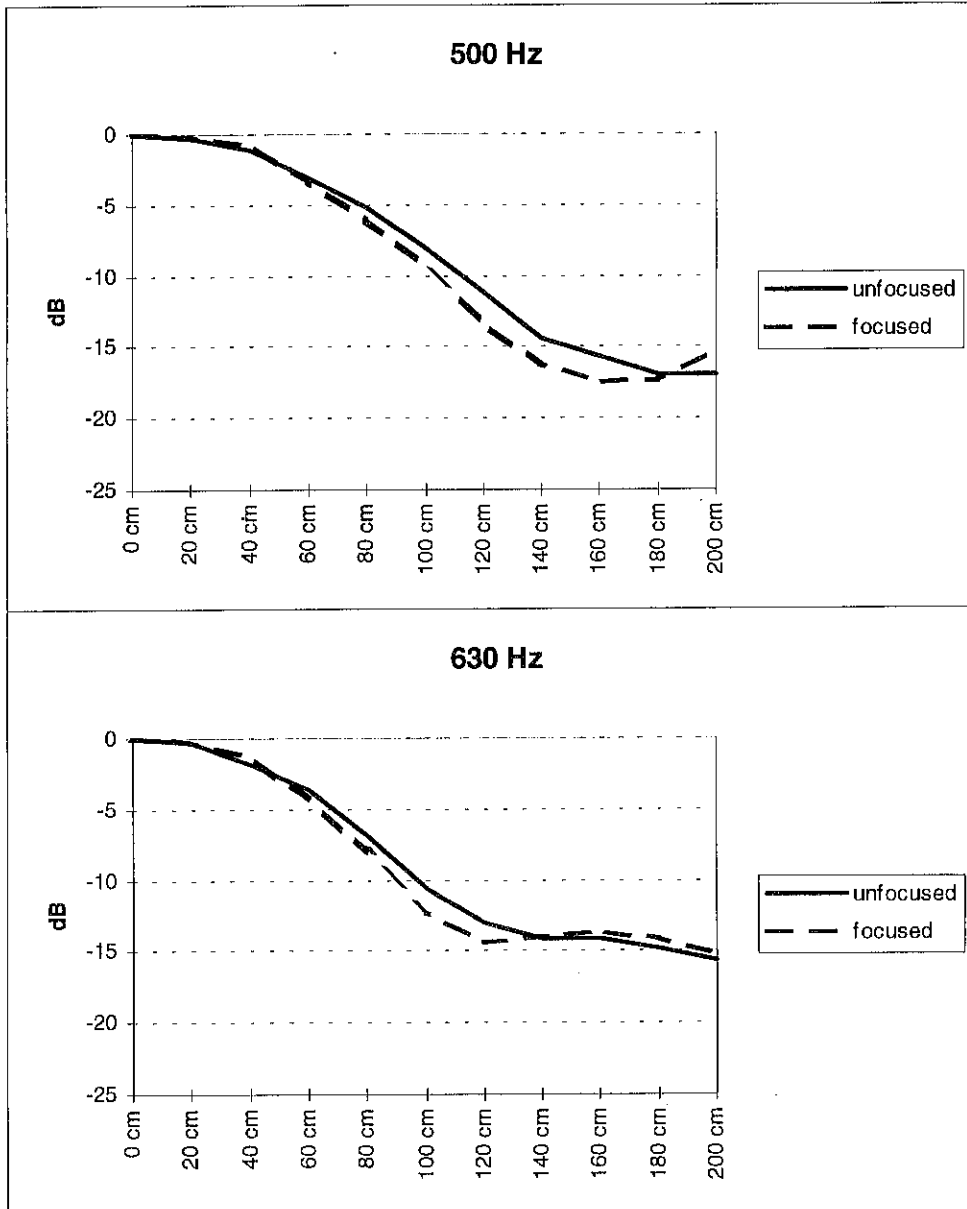


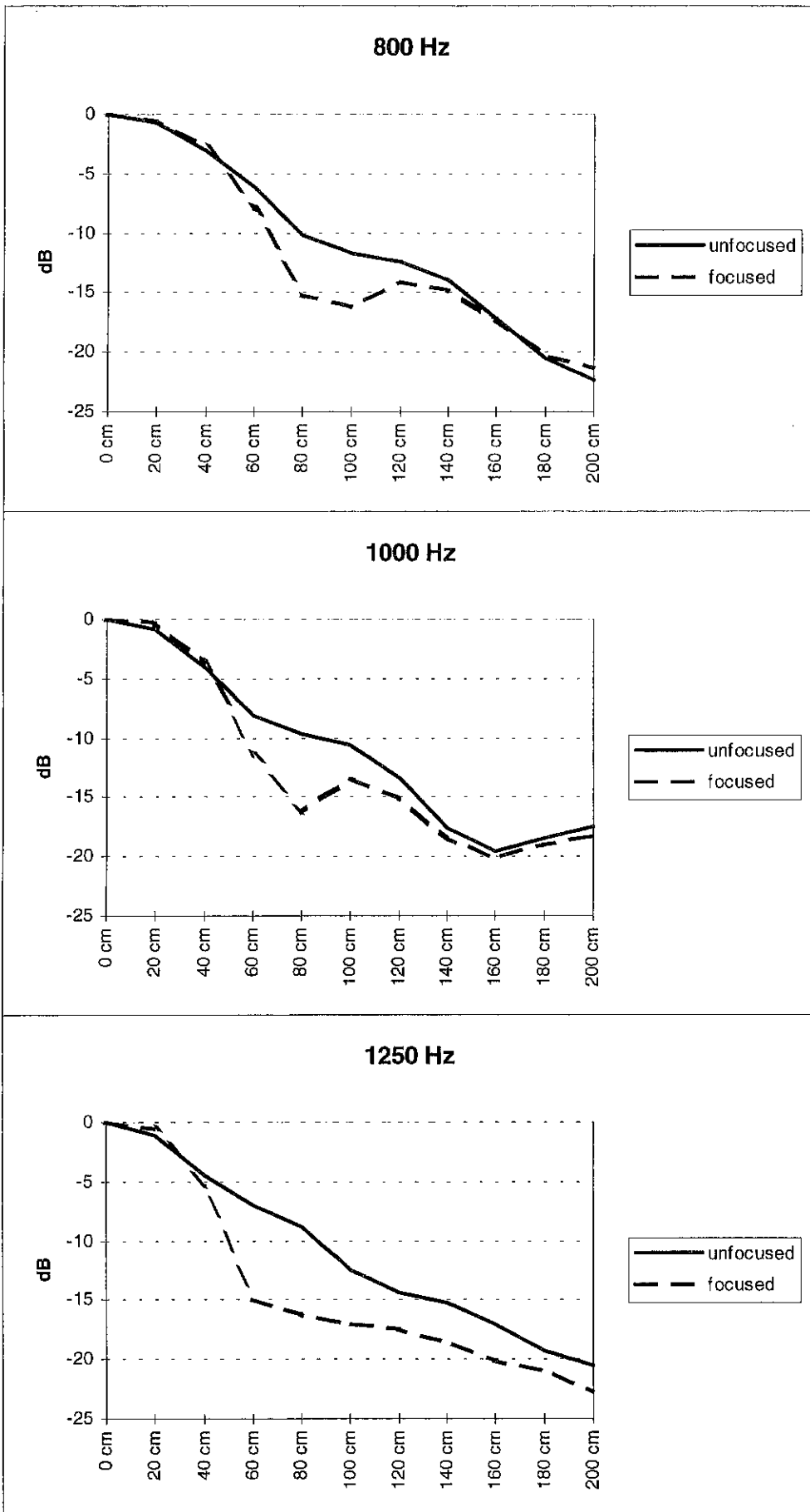


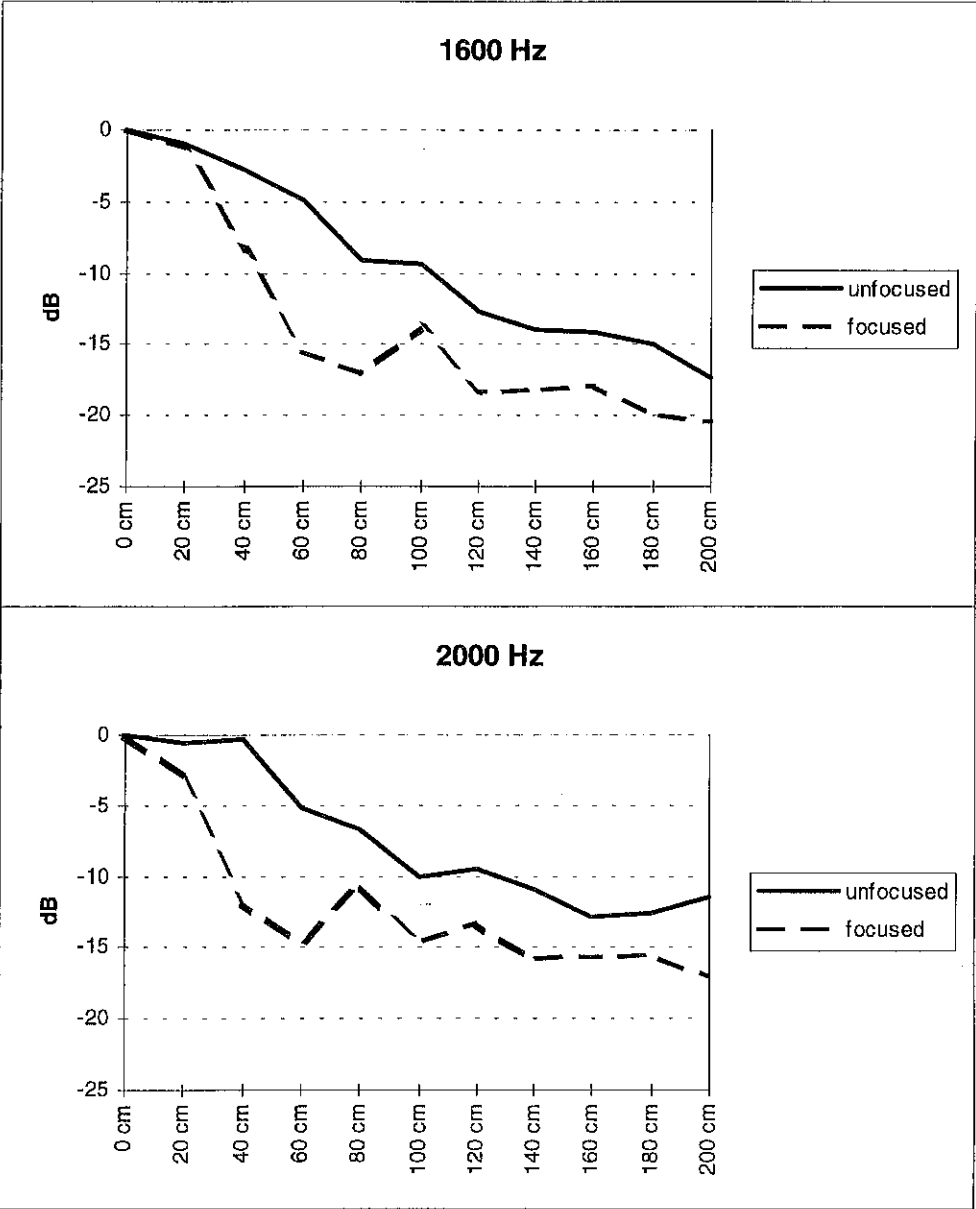


Annex B, Measurement in hemi-anechoic chamber

Measurement on a linear uniformly spaced array with eight transducers, both focused by displacement of the microphones, and unfocused. Se 2.2.2.







Annex C, Calculation of the array function

This annex contains MATLAB m-files for calculation of array functions. These files have been used to generate the figures in this text.

```
% This m-file calculates the so called Array Function
% at a given frequency and microphone positions.
%
% [G,fi]=AF(w,x,f);
%
%           w           Weighting coefficients in a column vector
%           x           Microphone position, se below
%           f           Frequency
%
%           G           The array function
%           fi          The corresponding angle fi
%
%
%           x=[ [x1 y1 z1]' [x2 y2 z2]' ... ]
%
function [G,fi]=AF(w,x,f);
c=340;
omega=2*pi*f;
lam=c./f;
N=length(w);
psi=0;

           uppl=360;

fi=(-uppl:uppl)*pi/2/uppl;

k=2*pi/lam*[sin(fi)*cos(psi);cos(fi)*cos(psi);sin(psi)*(-uppl:uppl)];
a=exp(-j*k*x).';
G=1/N.*w*a;

%G=20*log10(abs(G));
%G=20*log10(abs(G)/max(abs(G)));
```

```
% This m-file calculates the so called Array Function
% at a given frequency and microphone positions.
%
% [G,fi,psi]=AF2d(w,x,f);
%
%           w           Weighting coefficients in a column vector
%           x           Microphone position, se below
%           f           Frequency
%
%           G           The array function
%           fi          The corresponding angle fi
%           psi          and psi
%
%
%           x=[ [x1 y1 z1]' [x2 y2 z2]' ... ]
%
function [G,fi,psi]=AF2d(w,x,f);
c=340;
omega=2*pi*f;
lam=c./f;
N=length(w);
```

```

        uppl=45;% <----- here you choose the angle increment
                    % 45 corresponds to 2 deg, 90 to 1 deg

fi=(-uppl:uppl)*pi/2/uppl;

npsi=0;
for psi=-pi/4:pi./2/uppl:pi/4;
    npsi=npsi+1;
    k=2*pi/lam*[sin(fi).*cos(psi);cos(fi).*cos(psi);sin(psi).*ones(size(fi))];
    a=exp(-j*k'*x);
    G(npsi,:)=1/N.*w'*a;
end

psi=-pi/4:pi./2/uppl:pi/4;

%G=20*log10(abs(G));
%G=20*log10(abs(G)/max(max(abs(G))));

-----
% Design of an array with uniform/logarithmic spacing
%
% [x,d]=vdwaal(fmax,alphamax,N,M);
%
% source: Design of Logarithmically Spaced Constant-Directivity
% transducer Arrays, J. Audio Eng. Soc., vol 44, No 6 1996 June
%
%         fmax      Maximum measurable frequency
%         alphamax  Maximum angle deflection and grating lobe free area
%         N          The number of uniformly spaced microphones
%         M          Total number of microphones
%
function [x,d]=vdwaal(fmax,alphamax,N,M);

c=340;
d=c/(2*fmax*sin(alphamax*pi/180));

x=1:N;
x=d*(x-(N/2+.5));          %OBS N must be even

xa=zeros(1,(N-M)/2);
xa(1,1)=x(N)*N/(N-2);
for i=2:((M-N)/2)
    xa(i)=xa(i-1)*N/(N-2);
end;
size(xa);

x=[-fliplr(xa) x xa];
x(2,:)=zeros(1,M);
x(3,:)=zeros(1,M);

-----
% Calculation of weighting coeff. for the van der Waal array
%
% w=vdwaalw(d,fmax,N,x,f);
%
% source: Design of Logarithmically Spaced Constant-Directivity
% transducer Arrays, J. Audio Eng. Soc., vol 44, No 6 1996 June
%
%         fmax      Maximum measurable frequency
%         d          Distance between the microphones (calc. with vdwaal.m)
%         N          The number of uniformly spaced microphones

```

```

%           x           Microphone positions, (calc. with vdwaal.m)
%           f           Frequency
%
% OBS! You normally use a filter to obtain the weighting coeff. since
% they are frequency dependent. For calculations however, this method
% is a good alternative.
%
function w=vdwaalw(d,fmax,N,x,f);

[dum,M]=size(x);
for n=1:M
    S(n)=2/(N-1)*abs(x(1,n));
    if S(n)<d
        S(n)=d;
    end
end

for n=1:M
    if abs(x(1,n))>((N-1)*0.6997*343/f/2)
        T(n)=0;
    else
        T(n)=S(n).*5*(1+cos(2*pi*x(1,n)/((N-1)*0.6997*343/f)));
    end
end

w=(T/sum(T))';
%w=(w/max(w));

```

```

% This m-file calculates the weighting coeff. for the
% signal to noise ratio SNR dB for a eight microphone
% Dolph array.

```

```

%
% function w=dolphw(SNR);
%
function w=dolphw(SNR);

```

```

R=10^(SNR/20);
Zo=cosh(1/7*acosh(R))
a(4)=Zo^7;
a(3)=(-112*(Zo^5)+112*a(4))/16;
a(2)=(56*(Zo^3)-56*a(4)+20*a(3))/4;
a(1)=-7*Zo+3*a(2)-5*a(3)+7*a(4);
b=flipr(a);
w=[b a]';

```